ENSF 380

TOPIC 09: SCOPE

Use the Smallest Possible Scope for Variables

Acceptable Style

```
public class Example
{
    public void increment() {
        int base = 6;
        for (int i = 0; i < 5; i++) {
            int temp = base + i;
            System.out.println(temp);
        }
    }

    public static void main(String args[])
    {
        Example test = new Example();
        test.increment();
    }
}
```

01_Scoping

Unacceptable Style

```
public class Example
{
    int base = 6;
    public void increment() {
        int temp;
        for (int i = 0; i < 5; i++) {
            temp = base + i;
            System.out.println(temp);
        }
    }

    public static void main(String args[])
    {
        Example test = new Example();
        test.increment();
    }
}
```

Exercise 9.1

TEST YOUR UNDERSTANDING BY COMPLETING THE EXERCISE

Static vs Final

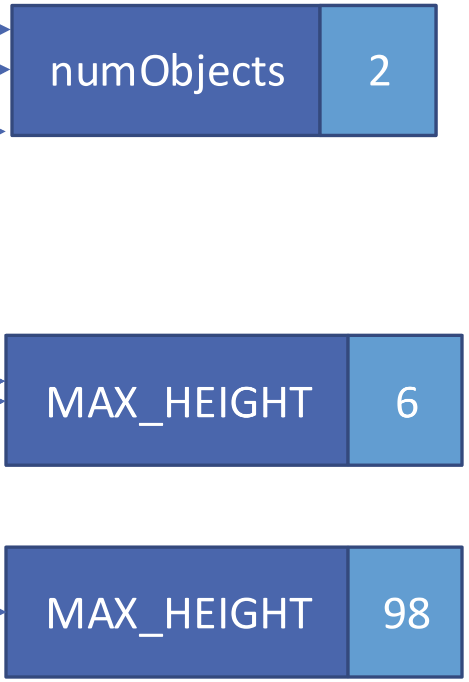
Tip: Final variables can be assigned values in the constructor, as well as at instantiation.

04_StaticVsFinal

static variables are class variables

FINAL variables retain the same value

```
…
public final int MAX_HEIGHT;
public static int numObjects;

public Example(int height) {
    this.MAX_HEIGHT = height;
    numObjects++;
}

public static void main(String[] args) {
    var myObj1 = new Example(6);
    System.out.println("Obj1 - Height: " + myObj1.MAX_HEIGHT +
       " NumObj: " + myObj1.numObjects);

    var myObj2 = new Example(98);
    System.out.println("Obj2 - Height: " + myObj2.MAX_HEIGHT +
       " NumObj: " + Example.numObjects);

    System.out.println("Obj1 - Height: " + myObj1.MAX_HEIGHT +
       " NumObj: " + numObjects);
}
…
```



Static vs Final

04_StaticVsFinal

...

```java
public final int MAX_HEIGHT;
public static int numObjects;

public Example(int height) {
    this.MAX_HEIGHT = height;
    numObjects++;
}

public static void main(String[] args) {
    var myObj1 = new Example(6);
    System.out.println("Obj1 - Height: " + myObj1.MAX_HEIGHT +
       " NumObj: " + myObj1.numObjects);

    var myObj2 = new Example(98);
    System.out.println("Obj2 - Height: " + myObj2.MAX_HEIGHT +
       " NumObj: " + Example.numObjects);

    System.out.println("Obj1 - Height: " + myObj1.MAX_HEIGHT +
       " NumObj: " + numObjects);
}
```

...

Terminal — -tcsh — 54x15

```
OOP> java edu.ucalgary.oop.Example
Obj1 - Height: 6 NumObj: 1
Obj2 - Height: 98 NumObj: 2
Obj1 - Height: 6 NumObj: 2
OOP>
```

Static Methods

08_Arrays/
01_SimpleArrays

+

...

...

```
public static int[] arrayFourElements() {
        // Create an array of integers containing the values shown
        int[] implicitLengthArray = {1, 2, 3, 5};
        System.out.println("implicitLengthArray is an int array with 4 
elements: ");
        // Iterate through the array and output the value.
        for (int tmpValue : implicitLengthArray) {
          System.out.println(tmpValue);
        }
        return implicitLengthArray;
}

public static void main(String[] args) {
        // int arrays
        int[] implicitLengthArray = arrayFourElements();
}
```

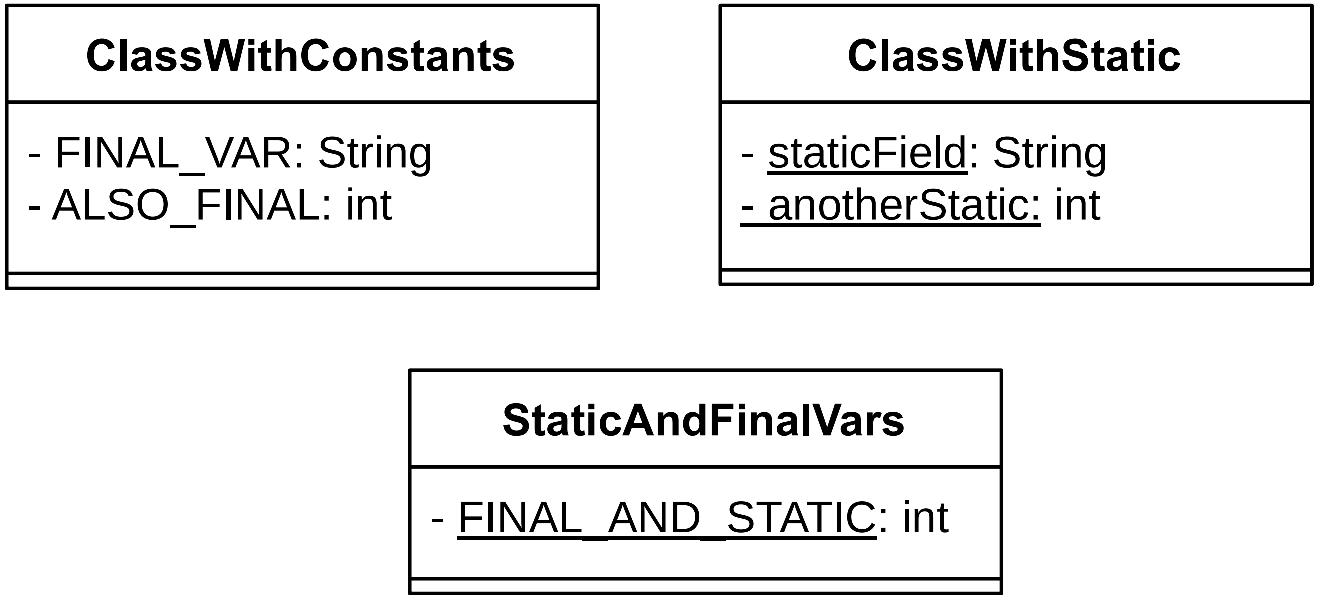
Global Variables

05_GlobalScope

```
public static final boolean thisWillNotWork = true;

public class Example {
    public final int MAX_HEIGHT;
    public static int numObjects;

    public Example(int height) {
        // set maximum height for this object
        this.MAX_HEIGHT = height;

        // Increment numObjects every time an object is instantiated
        numObjects++;
    }
```

…

Static and Final in UML



Summary: Knowledge

Scoping

Static

Final